Assignment of FODS

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# 1. Arithmetic Operations Function: Write a program to create a function that
accepts
# two numbers and displays their result for addition, subtraction, and
multiplication.
def arithmetic_operations(num1, num2):
    print("Addition =", num1 + num2)
    print("Subtraction =", num1 - num2)
    print("Multiplication =", num1 * num2)
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
arithmetic_operations(a, b)

# 2. Geometry Calculator: Write a program to implement separate functions for
the
# following calculations. Each function should take necessary user inputs
(like radius or
# side length) as parameters and return the result. 10 marks
# a. Area of a Circle
# b. Perimeter of a Square
# c. Area of a Triangle
# d. Volume of a Cube

import math

# Area of Circle
def area_circle(radius):
    return math.pi * radius * radius

# Perimeter of Square
def perimeter_square(side):
    return 4 * side

# Area of Triangle
def area_triangle(base, height):
    return 0.5 * base * height

# Volume of Cube
def volume_cube(side):
    return side ** 3

# User input
r = float(input("Enter radius of circle: "))
s = float(input("Enter side of square/cube: "))
b = float(input("Enter base of triangle: "))
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h = float(input("Enter height of triangle: "))

# Display results
print("Area of Circle =", area_circle(r))
print("Perimeter of Square =", perimeter_square(s))
print("Area of Triangle =", area_triangle(b, h))
print("Volume of Cube =", volume_cube(s))

# 3. Palindrome Checker: Write a program to create a function that accepts one string as
# input and checks whether it is a palindrome (reads the same forwards and backwards)
# or not. 10 marks
# Function to check palindrome

def palindrome_check(text):
    if text == text[::-1]:
        return "Palindrome"
    else:
        return "Not a Palindrome"

word = input("Enter a string: ")
print(palindrome_check(word))

# 4. Word Frequency Counter: Write a function that takes a list of words as input and
# returns a dictionary showing the frequency of each word. Test this with a list
# containing at least five words, some of which should repeat.

def word_frequency(words):
    frequency = {}

    for word in words:
        if word in frequency:
            frequency[word] += 1
        else:
            frequency[word] = 1

    return frequency
word_list = ["apple", "banana", "apple", "orange", "banana", "apple"]
print(word_frequency(word_list))

# 5. Grade Sorter: Write a function that accepts a list of student grades (integers) and
# returns the grades sorted in descending order. 10 marks
# Function to sort grades in descending order

def sort_grades(grades):
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    return sorted(grades, reverse=True)
grades = [78, 45, 90, 67, 88]
print("Sorted Grades:", sort_grades(grades))

# 6. Search and Identify: Write a function that accepts two parameters: a list
of fruit
# names and a specific fruit to search for. The function should return the
index of the
# fruit if found; otherwise, it should return &quot;Item not in list.&quot; 10
marks
# Function to search fruit in list

def search_fruit(fruits, item):
    if item in fruits:
        return fruits.index(item)
    else:
        return "Item not in list."
fruit_list = ["apple", "banana", "orange", "mango"]
search_item = input("Enter fruit to search: ")
print(search_fruit(fruit_list, search_item))

# 7. Vowel Counter: Write a function count_vowels(text) that takes a string as
input and
# returns a dictionary showing how many times each vowel (a, e, i, o, u)
appears,
# ignoring case.

def count_vowels(text):
    vowels = {'a':0, 'e':0, 'i':0, 'o':0, 'u':0}

    for char in text.lower():
        if char in vowels:
            vowels[char] += 1

    return vowels
sentence = input("Enter text: ")
print(count_vowels(sentence))

# 8. Unique Element Merger: Write a program to accept two different lists of
numbers
# from the user. Return a new list that contains only the unique elements from
both lists
# combined, sorted in ascending order. 10 marks
# Input lists from user
list1 = list(map(int, input("Enter first list numbers separated by space:
").split()))
list2 = list(map(int, input("Enter second list numbers separated by space:
").split()))
unique_list = sorted(list(set(list1 + list2)))
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print("Unique Sorted List:", unique_list)


# 9. Library Record System: Write a program to ask for the details of 3 books
(Title,
# Author, ISBN, Price). Store these in a dictionary and display the
information in a
# clear, well-formatted table or list.

library = {}
for i in range(1, 4):
   print(f"\nEnter details for Book {i}")

   title = input("Title: ")
   author = input("Author: ")
   isbn = input("ISBN: ")
   price = float(input("Price: "))

   library[title] = {
       "Author": author,
       "ISBN": isbn,
       "Price": price
   }
print("\nLibrary Records")
print("-" * 40)

for title, details in library.items():
   print("Title :", title)
   print("Author:", details["Author"])
   print("ISBN  :", details["ISBN"])
   print("Price :", details["Price"])
   print("-" * 40)


# 10. Number Guessing Game: Create a program to design a number guessing game
with
# the following specifications 5:
# a. The program generates a random number between 1 and 10.
# b. The player has three attempts to guess the number.
# c. If the guess is correct, display &quot;Congratulations! You won!&quot;
# d. If the guess is higher than the random number, display &quot;Too
High.&quot;
# e. If the guess is lower, display &quot;Too Low.&quot;

import random

number = random.randint(1, 10)
attempts = 3
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print("Guess the number between 1 and 10")

for i in range(attempts):
   guess = int(input("Enter your guess: "))

   if guess == number:
       print("Congratulations! You won!")
       break

   elif guess > number:
       print("Too High.")

   else:
       print("Too Low.")
if guess != number:
   print("Game Over! The number was", number)
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By : Abhipsha Bhandari